

GENERAL SPECIFICATION G5.34

Low Voltage Switchgear

CONTENTS

G5.34 Low Voltage Switchgear1

 G5.34.1 General.....1

 G5.34.2 Products2

 G5.34.3 Execution.....18

G5.34 LOW VOLTAGE SWITCHGEAR

G5.34.1 General

(a) Introduction

- (i) Perform all work; design, supply, deliver and install all materials and equipment, and installation methods in accordance with, but not limited to, the latest applicable rules, regulations, requirements, national and international codes, as listed in General Specification G5.2 - Standards, Codes and Regulations.

(b) References

- (i) The following is a list of standards which may be referenced in this Section:

- 1) British Standards Institute (BSI): BS 5252, Framework for Colour Co-ordination for Building Purposes.
- 2) International Electrotechnical Commission (IEC)

Standard	Subject
IEC 61439-1	Low Voltage Switchgear and Controlgear Assemblies Part 1 Type-Tested and Partially Type-Tested Assemblies.
IEC 60529	Degrees of Protection Provided by Enclosures.
IEC 60947	Low Voltage Switchgear and Controlgear.
IEC 61439	Low-voltage switchgear and controlgear assemblies.
IEC TR 61641	Enclosed low-voltage switchgear and controlgear assemblies - Guide for testing under conditions of arcing due to internal fault.

- (ii) SPRING Singapore

Standard	Subject
CP 5	Code of Practice for electrical installations.
SS 551	Code of Practice for earthing.
SS 555	Code of Practice for protection against lightning.

- (iii) Conflicts, if any, that may exist between the above reference standards, codes and regulations will be resolved at the discretion of the S.O..

(c) Submittals

- (i) Shop Drawings, and Technical Data: As a minimum, submit the following items:
 - 1) Descriptive product information;
 - 2) Itemised Bill of Material;
 - 3) Dimensional drawings;

- 4) Operational description;
 - 5) Safe switching and operating procedure;
 - 6) Anchoring instructions and details;
 - 7) Single-line, three-line, and control schematic drawings / communication network schematics;
 - 8) Connection and interconnection drawings;
 - 9) Circuit Breakers: Copies of time-current characteristics;
 - 10) Earth Fault Protection: Relay time-current characteristics;
 - 11) Bus data;
 - 12) Incoming line section equipment data;
 - 13) Cable or busway entrance locations;
 - 14) Type Test certificate;
 - 15) Manufacturer's installation instructions;
 - 16) Certified Factory Test Report;
 - 17) Operation and Maintenance Manual; and
 - 18) Manufacturer's Certificate of Proper Installation.
- (d) Spares
- (i) Provide spare parts, tools and test equipment as specified in General Specification G2.28 - Spare Parts, Tools and Test Equipment.

G5.34.2 Products

- (a) Manufacturers
- (i) Qualifications
 - 1) Normally engaged in the manufacture of switchgears;
 - 2) Minimum of 10 years' experience in the design and manufacture of switchgear;
 - 3) Certified under ISO 9001 for manufacture of electrical apparatus;
 - 4) Fully Type Tested in accordance with IEC 61439; and
 - 5) Internal Arc Containment, tested in accordance with IEC 61641.
- (b) General
- (i) Service: 415/240 volts, three-phase, four-wire earthed wye connection, clearly marked to show the phases or identification by means of coloured plastic sleeves. Plastic tapes shall not be used.
 - (ii) Short circuit rating: The complete assembly shall have rated short-time withstand current of 65,000 amperes rms symmetrical for 3 seconds.
 - (iii) Fully Type Tested, Designed and Assembled in accordance with IEC 61439, IEC 60947, and IEC 61641.

- (iv) Arc Fault Containment Design: The LV Switchgears shall be designed and dimensioned to prevent arc fault from occurring. The switchgears shall be designed to limit the arc in the compartment where it occurs, so as to minimize the consequences in the event of arc occurrence. The switchgears shall be tested in accordance with IEC 61641 for personal safety (criteria 1 to 5) and plant & safety (Criteria 6 to 7).
- (v) Switchgear and its major components shall be end products of one manufacturer in order to achieve standardisation for appearance, operation and maintenance, spare parts replacement, and manufacturer's services.
- (vi) All LVSGs shall be from a single manufacturer and shall be procured directly from the manufacturer.
- (vii) LVSGs shall be of compact design and withdrawal type.
- (viii) Operating Conditions
 - 1) Ambient Temperature: Maximum 40 degrees C;
 - 2) Elevation: <1000m above Mean Sea Level; and
 - 3) Equipment shall be fully rated without derating for operating conditions.
- (ix) LVSG's shall monitoring occurring via a high-availability field communications network;
- (x) Lifting lugs on all equipment and devices weighing over 45 kg.
- (xi) Anchor Bolts: Type 316L stainless steel, sized by equipment manufacturer.
- (c) Stationary Structure
 - (i) Type: IEC 61439, Form 4b switchgear construction, consisting of pull section with cable terminations, metering, breaker, and auxiliary sections assembled to form a rigid, self-supporting, metal enclosed structure.
 - (ii) Material: Sheet steel, shall not less than 2.5 mm thick for panel front plates and 2.0 mm for panel side and rear plates and inner cubicle partitions.
 - (iii) Earthed metal barriers between each breaker, main bus, branch cabling, and instrumentation/control.
 - (iv) Modular-designed steel frame with removable plates and individual, bolted, steel-framed vertical sections.
 - (v) There shall be no danger of water penetrating into compartments housing active components. The clearance of active components from the switchroom floor shall be at least being 300 mm. The busbar compartment shall be at least 900 mm above switchroom floor. Minimum clearance of 1500 mm in front and 600 mm at rear and sides shall be provided for 415 V switchboards.
 - (vi) Individual, hinged doors over each breaker, metering, and auxiliary compartments.
 - (vii) Cable Installation and Termination Compartments
 - 1) Rear bolted cover plates or hinged doors, capable of being bolted closed;
 - 2) Cable bending space adequate for the cables being installed;

- 3) Cable supports in each vertical section; and
 - 4) All cables shall be bottom entry.
- (viii) Breaker Compartments
- 1) Individual, earthed compartments, with:
 - Sheet steel, top, bottom, sides, and ventilated compartment door with padlocking features;
 - Flame retardant, arc track resistant non-metallic rear barrier;
 - Drawout rails, stationary breaker contacts, interlocks, and necessary control and indicating devices;
 - Shutters over all stationary contacts when breaker is in either TEST or full DISCONNECT position; and
 - Padlocking provision on rackout rails for locking breaker in either TEST or DISCONNECT position.
 - 2) Drawout Mechanism
 - Shall retain removable element in connected position;
 - Mechanical interlocks to ensure breaker is open before moved from any position, or when between positions;
 - Four Distinct Breaker Positions: CONNECTED, TEST, DISCONNECTED, and WITHDRAW;
 - Indicators to display breaker position; and
 - Capable of being operated without opening breaker door.
 - 3) Breaker frame earthed to steel frame throughout travel of drawout mechanism;
 - 4) Each compartment designed for specific breaker frame size; and
 - 5) Future breaker compartments fully equipped with all electrical connections, bolted metal barrier across compartment face, and compartment door.
- (ix) Slide-Out Instrument Tray
- 1) Mount above associated breaker;
 - 2) Accessible from front of switchgear; and
 - 3) For all control circuitry, breaker close and trip fuses, indicating lights, feeder metering, and rotary phase selector switch.
- (x) Auxiliary sections equipped with devices shown on the Drawings or as specified, including Data Concentrators and Data Network Hardware, with hinged door over each compartment.

- (xi) Busway Entrance
 - 1) Located for busway entrance from top;
 - 2) Suitable for busway flange connected to internal bus work; and
 - 3) Busway size shall be as shown on SLD.
- (xii) Cable Entrance
 - 1) Located for cable entrance from bottom; and
 - 2) Suitable for the incoming cable size as shown on SLD.
- (d) Enclosure
 - (i) Finish: Baked enamel applied over a rust-inhibiting, phosphate base coating.
 - 1) Colour
 - Exterior: Grey, RAL 7032, finish;
 - Interior: Manufacturer's standard; and
 - Unpainted Parts: Plated for corrosion resistance.
 - (ii) Indoor Enclosure
 - 1) IEC 60529, Type IP 43 with formed edges on hinged and non-hinged panels;
 - 2) Rear access, full-height, bolt-on panels for each enclosure section; and
 - 3) Cable Termination Access: Padlock provision.
 - (iii) Outdoor Enclosure
 - 1) IEC 60529, Type IP 65;
 - 2) The cubicle shall be fitted with lockable outer door;
 - 3) The whole assembly shall be weather proof and vandal proof;
 - 4) Inner cubicles shall also be weather proof to allow operation of controls when the outer door is open during inclement weather;
 - 5) Door mounted components shall be protected by vandal resistant secondary glazed door so that all controls and indicators are clearly visible;
 - 6) The cubicles shall have a weather canopy roofing sloped to the rear and over hand the front by at least 300 mm; and
 - 7) Cable Termination Access: Shall have Padlock provision.
- (e) Buswork
 - (i) Material: Phase isolated and insulated tin-plated copper throughout entire length of sufficient cross section to limit current density to 1.5 amperes per square mm. The short time current rating shall be compatible to that of breaker.
 - (ii) Bolts and nuts for the busbar connection shall be copper alloy or non-magnetic type with thermal co-efficient of expansion close that of copper and shall be able to withstand maximum force exerted during the electrical fault. Bolts and nuts shall be fitted with spring washer and lock nuts.

- (iii) The busbars shall be adequately sized with calculations to substantiate the performance rating. The short time current rating shall be compatible to the connected breakers. The busbar size shall provide rated current with less than 30 deg C, temperature rise over the ambient temperature.
- (iv) Brace for short circuit currents 85,000 amperes rms symmetrical 3 seconds.
- (v) Main Horizontal Bus: Non-tapered, continuous current rating as shown on the Drawings.
- (vi) Neutral Bus: Continuous current rating 100 percent of main horizontal bus rating.
- (vii) Earth Bus
 - 1) Material: Tinned Copper;
 - 2) Rating: Continuous current rating 100 percent of main horizontal bus rating;
 - 3) Bolted to each vertical section;
 - 4) Earth lug for 240 mm² copper conductor on each end of bus; and
 - 5) Bus Connections and Joints: Bolted.
- (viii) Extend each bus entire length of switchgear with provision for extension to future units without need for adapter panels.
- (f) Protective Devices
 - (i) Power Air Circuit Breakers
 - 1) Main, bus connector, and feeder breakers 800 amperes frame and larger;
 - 2) Power circuit breakers shall meet all applicable requirements of IEC 60947-2;
 - 3) Arrangement: Selective Type-2 co-ordinated main with zone-selective and fully rated branch feeder circuit breakers;
 - 4) Interlock: Mechanical and electrical interlock shall be provided between main incoming breakers and bus coupler breaker;
 - 5) Electrically and mechanically trip-free with:
 - Four poles for main and tie circuit breakers;
 - Four poles for feeder circuit breakers unless indicated otherwise;
 - Self-aligning primary and secondary contacts;
 - Integral, solid state, over-current trip programmer;
 - Arc quenchers;
 - Closing Mechanism: Electrical;
 - Stored energy mechanism with maximum five-cycle closing;
 - Electronic Trip Units; and
 - Shunt trip device for all breakers.

- 6) Individually mounted, drawout breaker listed for 100 percent continuous ampere rating;
 - 7) Current Rating: As shown on the Drawings;
 - 8) Interrupting Rating: 65,000 amperes rms symmetrical (Ics) 3 seconds at 415 volts;
 - 9) Integral operational current display;
 - 10) Phase failure, phase reversal relay to trip circuit breakers through shunt trip device on main breakers;
 - 11) Pad lock facility; and
 - 12) Communications Interface: Integral Ethernet interface, protocol to suit the Power Monitoring System. Status data to be transmitted includes Open, Closed, Tripped.
- (ii) Moulded-Case Circuit Breakers
- 1) Feeder circuit breakers less than 800 amperes frame;
 - 2) Individually mounted, withdrawable;
 - 3) Arrangement: Selective Type-2 co-ordinated main with fully rated feeder circuit breakers;
 - 4) Circuit breakers shall meet all applicable requirements of IEC 60947-2;
 - 5) Interrupting Rating: 65,000 amperes rms symmetrical (Ics) 3 seconds at rated voltage of 415 volts;
 - 6) Current Rating: As shown on the Drawings;
 - 7) Mechanical interlock to prevent opening compartment door while breaker is in closed position;
 - 8) Accessories: Electronic Trip unit with adjustable overload, overcurrent, Earth fault relay and shunt trip. Each MCCB shall have necessary auxiliary contacts for monitoring of ON, OFF and TRIP status; and
 - 9) Pad lock facility.
 - 10) Monitoring Facilities: Status contacts connected to PMS i/o module within LVSG.
- (iii) Mechanical Operation of Power Air Circuit Breakers
- 1) Front mounted, spring charging handle;
 - 2) Mechanical closing escutcheon mounted pushbutton; and
 - 3) Mechanical trip, escutcheon mounted, trip pushbutton handle.
- (iv) Electrical Operation for Main and Tie Circuit Breakers
- 1) Automatic spring charging by motor, plus manual charging;
 - 2) Electrically closing, escutcheon mounted, pushbutton with mechanical closing upon loss of control power;

- 3) Electrical trip, escutcheon mounted, trip pushbutton; and
- 4) Control Power Voltage: 230 AC.
- (v) Colour-Coded Visual Indicators: Contacts OPEN and CLOSED, plus mechanism CHARGED and DISCHARGED.
- (vi) Accessories
 - 1) Slow breaker closing handle for contact adjustments;
 - 2) Breaker lifting hoist;
 - 3) Two normally open and two normally closed auxiliary contacts, in addition to those required for circuit breaker operation, on main, tie, and feeder breakers; and
 - 4) Shunt trip for main, tie and feeder breakers.
- (vii) Test Facilities
 - 1) Breakers with integral external test points for portable test kit;
 - 2) Handheld test kit for functional testing of trip circuitry of each breaker; and
 - 3) Separately mounted panel to test earth fault by not tripping breaker.
- (viii) Electronic Trip Units: Flux-shift trip and current sensors:
 - 1) Protective Programmers:
 - Self-powered, automatic rms sensing micro-electronic processor;
 - No external relays or accessories;
 - Printed circuit cards with gold-plated contacts;
 - Programmable Controls:
 - Fixed-point, with repetitive accuracy and precise unit settings; and
 - Trip adjustments made by non-removable, discrete step switching.
 - Field-Installable Rating Plugs:
 - Long-time pickup LED indicator and test receptacle;
 - Matching load and cable requirements;
 - Interlocked with tripping mechanism;
 - Breaker to remain trip-free with plug removed; and
 - Keyed rating plugs to prevent incorrect application.
 - Long-time pickup light;
 - Selective co-ordination time/current curve shaping adjustable functions:
 - Current setting;
 - Long-time pickup;
 - Long-time delay;

- Instantaneous pickup with short-time for main tie and feeders;
 - Short-time pickup for main tie and feeders;
 - Short-time delay for main tie and feeders with I²T function, and IN-OUT switch;
 - Earth fault pickup;
 - Earth fault delay with I²T function;
 - High instantaneous pickup with short-time delay;
 - Zone selective interlock; and
 - External earth fault relay shall be provided where required to meet the selective co-ordination.
- Fault Trip Indicators: Mechanical push-to-reset type for overload and short circuit overload plus earth fault trip; and
- Rejection Pins: For each programmer frame size.
- 2) Phase Current Sensors:
 - Single ratio type;
 - Fixed, mounted on breaker frame;
 - Moulded epoxy construction; and
 - One toroidal type for each phase.
- 3) Earth Fault Sensor:
 - Neutral bar single-ratio CT mounted in cable compartment;
 - Moulded epoxy construction; and
 - Shorting bar.
- 4) Portable Test Set: AC/DC static, full function unit for checking programmer's time-current characteristics of programmer.
- (g) Automatic Transfer Control (ATC) Equipment
 - (i) Where shown, provide ATC equipment to manually and automatically transfer switchgear loads from a preferred power source (switchgear main breaker) to an alternate power source (as shown on single-line diagrams). The ATC equipment shall provide positive interlocking between the closing circuits of the motor operated main circuit breakers in both the AUTOMATIC and MANUAL mode of operation to prevent paralleling the sources (only one of the two closed at one time).

- (ii) The ATC equipment shall include a control relay panel or programmable logic controller (PLC) and ATC operator's control panel located in a completely barriered compartment included in the switchgear line-up. Relay panel (or PLC) shall include all relays, timers, control fuses, control power transformers, and terminal board for interconnecting wiring to each source breaker. All wiring shall be as specified herein. ATC operator's control panel shall include indicating lights and selector switches with nameplates describing the functions of the devices. Indicating lights and selector switches shall be as specified under General Specification G5- Basic Electrical Materials and Methods. Provide ATC operating instruction plate. Locate the plate above the operator's control panel. Submit instruction plate for review before engraving. Letter height shall be not less than 4 mm. The ATC shall include the following minimum features. Additional features not specified but required to provide a complete and operable system shall also be provided:
- 1) Three-phase voltage sensing on both sources. Voltage relays pickup and dropout adjustable 70 to 100 percent rated voltage. Field set as instructed by the S.O.;
 - 2) Delayed transfer to healthy source 0 to 3 minutes, adjustable, to allow switchgear load to shut down;
 - 3) MANUAL/AUTOMATIC ATC mode selector switch, key operated;
 - 4) MANUAL/AUTOMATIC retransfer selector switch, key operated;
 - 5) Source selector switch, key operated;
 - 6) Test switches to simulate loss of source to each main breaker;
 - 7) Main circuit breaker overcurrent trip lockout of ATC;
 - 8) Main circuit breaker control switches for manual operation;
 - 9) Indicating lights as follows:
 - Circuit breakers open (red) and closed (green);
 - Circuit breakers tripped (amber);
 - Source power available for each main breaker (clear); and
 - ATC Manual Operating Modes: AUTO (orange), MANUAL (blue).
 - 10) Delayed automatic retransfer 0- to 30-minute timer, adjustable to provide verification of preferred source before ATC initiation;
 - 11) Delay in neutral position, 0 to 3 minutes, to allow switchgear loads to shut down;
 - 12) AC close and trip circuit breaker control power with control power transfer relay;
 - 13) Alarm Form C contacts rated 250-volts, 5 amperes for indicating ATC lockout;
 - 14) ATC position auxiliary contacts, rated 250-volts, 5 amperes. Provide Form C contacts for each main breaker for interfacing with the monitoring and control system; and

- 15) Form C contacts rated 5 amperes, 230 volts for remote alarm of ATC system trouble.
- (iii) ATC operation with selector switches Items 3 and 4 above in the AUTO position, preferred main breaker closed, alternate main breaker open shall be as follows:
 - 1) Step 1: After preferred source fails for more than 20 seconds, adjustable per Item 2 above, main breaker with failed source is opened;
 - 2) Step 2: If the preferred source does not return, at the end a pre-set delay period of the 0 to 5 seconds timer, the alternate source breaker shall be closed; and
 - 3) When the failed source returns, a delayed automatic retransfer 0- to 30-minute timer starts timing. At the end of the pre-set delay of the 0- to 30-minute timer, the alternate source breaker is opened and a 0- to 30-second timer starts timing. At the end of pre-set delay period of 0- to 30-second timer, the preferred main breaker is closed.
- (iv) ATC operation with MANUAL/AUTOMATIC mode selector switch in AUTO position, MANUAL/AUTOMATIC retransfer selector switch is in MANUAL position, preferred main breaker closed and alternate main breaker open:
 - 1) Automatic transfer of switchgear load from preferred source to alternate source shall be as described in Steps 1 and 2 above for complete automatic operation and Step 3 above shall not be initiated. However, if the alternate source fails, switchgear loads shall be transferred to the preferred source; Step 3 above is initiated.
 - 2) Manual retransfer after preferred source has been restored shall be possible with the ATC mode selector switch in the MANUAL position by operating the circuit breaker manual control switches or remotely via the plant MCS.
- (v) With MANUAL/AUTOMATIC mode selector switch in MANUAL position it shall be possible to OPEN the preferred main breaker and close the alternate main breaker. It shall not be possible for both main breakers to be closed at same time.
- (vi) Communications System Interface: Each ATC shall be capable of communicating to the PMS over an Ethernet network:
 - 1) Media Connection: Fibre optic;
 - 2) Protocol: ModBus TCP/IP;
 - 3) Network Addressing: Co-ordinated through S.O., entered and verified after units are in place and operational; and
 - 4) Data Registers: As a minimum, provide the following information as data which can be qualified by the ModBus host:
 - All alarms;
 - OPEN/CLOSE status of main circuit breakers;
 - AUTO status of ATC; and
 - Retransfer control in manual mode.

- (h) Control Wiring
 - (i) Single-conductor, Class B, stranded copper;
 - (ii) Insulation: Flame retardant, thermoset, rated 600 volts for control, instrumentation, and power/current circuits;
 - (iii) Shielded cable rated 600 volts for transducer output and analogue circuits;
 - (iv) Enclosed in top and vertical steel wiring troughs, and front-to-rear in non-metallic wiring troughs;
 - (v) Conductor Lugs: Pre-insulated, self-locking, spade type, with reinforced sleeves;
 - (vi) Identification: Individually, with permanent wire markers at each end; and
 - (vii) Splices: Not permitted in switchgear wiring.
- (i) Terminal Blocks
 - (i) Enclosed in steel wiring troughs;
 - (ii) Rated 600 volts, 30 amperes minimum, one-piece barrier type with strap screws;
 - (iii) Shorting type for current transformer leads;
 - (iv) Provide terminal blocks for:
 - 1) Conductors connecting to circuits external to switchgear;
 - 2) Internal circuits crossing shipping splits; and
 - 3) Equipment parts requiring replacement and maintenance.
 - (v) Spare Terminals: Not less than 20 percent;
 - (vi) Group terminal blocks for external circuit wiring leads;
 - (vii) Maintain 150 mm minimum space between columns of terminal blocks; and
 - (viii) Identification: Permanent, for each terminal and columns of terminals blocks.
- (j) Instrumentation and Metering
 - (i) Potential Transformers
 - 1) Type: Moulded;
 - 2) Accuracy Classification: Class 1 at burden imposed by metres and instruments, including future;
 - 3) Primary Fuses: Current-limiting; and
 - 4) Secondary Fuses: Current-limiting.
 - (ii) Current Transformer
 - 1) Type: Moulded donut;
 - 2) Accuracy: Class 5P20 for protection, Class X for differential protection, and Class 0.5 at burden imposed by metres and instruments;
 - 3) Shorting type terminal boards for current transformer leads; and

- 4) Where multi-ratio current transformers are specified, a label shall be provided, clearly indicating the connections required for the alternative ratios. These connections shall be shown on the panel wiring diagrams.
- (k) Protection Relay (Type PR-5)
 - (i) All Protection Relays shall be from the same manufacturer as switch gear unless approved by S.O.;
 - (ii) External protection relays shall also be provided for all Incoming and TIE feeders;
 - (iii) Protection relays as minimum shall have following protections: 50, 50N, 51, 51N, 64 and 27;
 - (iv) Relay contacts shall be capable of making and breaking the maximum current which can flow in the respective circuit, they control. Contacts for remote alarms and indications shall be volt free;
 - (v) Provide front panel accessible port connection for local interrogation and programming using a portable PC (by others) and a Windows-based program specific to this unit. Program to enable full access to unit's data and provide, as a minimum, the following features:
 - 1) View actual values;
 - 2) View unit status;
 - 3) View/edit settings (online and offline);
 - 4) View event record (for troubleshooting);
 - 5) Help screens and step-by-step tutorial; and
 - 6) Provision for installation of firmware upgrade(s) to unit.
- (l) Energy Meter (Type EM-2) for incomers and outgoing feeders.
 - (i) All Energy Meters shall be from the same manufacturer as switch gear unless approved by S.O.;
 - (ii) Solid state device with easy to read displays;
 - (iii) Direct voltage input up to 600 volts AC;
 - (iv) Current input via current transformer with 5 amperes secondary;
 - (v) Programmable current and potential transformer ratios;
 - (vi) Programmable limits to activate up to four alarms;
 - (vii) Selectable Voltage Measurements: Line-to-line or line-to-neutral and wye or delta;
 - (viii) Simultaneous Display
 - 1) Volts, three-phase;
 - 2) Amperes, three-phase;
 - 3) Kilowatts, Reactive KVA, and KVA;
 - 4) Kilowatt-hours;

- 5) Power factor;
 - 6) Frequency; and
 - 7) Earth leakage mA.
- (ix) Power Quality Monitoring
- 1) Individual and total harmonics up to 50th harmonic;
 - 2) Simultaneous capture of events on all channels:
 - Up to 60-cycles (minimum) each; and
 - Resolution adjustable up to 64 samples per cycles, minimum.
 - 3) Sag/Swell Monitoring:
 - Record magnitude, duration and date stamp.
- (x) Communications System Interface: Each power meter shall be capable of communicating to a host computer system over an Ethernet network:
- 1) Media Connection: Copper (within LVSG);
 - 2) Protocol: ModBus TCP/IP or other protocol to suit Power Monitoring System. Refer to Particular Specifications P6 ICA and Drawings;
 - 3) Network Addressing: Co-ordinated through S.O., entered and verified after units are in place and operational;
 - 4) Data Registers: As a minimum, provide the following information as data which can be queried by the host:
 - All items listed for display; and
 - All alarms.
 - 5) All power measurement and power quality monitoring data shall be sent to host computer / SCADA / PMS / EMS system as per Specifications and Drawings.
 - 6) Provide front panel accessible port connection for local interrogation and programming using a portable PC (by others) and a Windows-based program specific to this unit. Program to enable full access to unit's data and provide, as a minimum, the following features:
 - View actual values;
 - View unit status;
 - View/edit settings (online and offline);
 - View event record (for troubleshooting);
 - Help screens and step-by-step tutorial; and
 - Provision for installation of firmware upgrade(s) to unit.

- (m) LVSG Monitoring Facilities
 - (i) The status of all incoming, TIE, and feeder circuit breakers, and ATC's, shall be connected to the PMS. Provide all necessary i/o units and cabling works for the complete system, including daisy-chain Ethernet cabling and media converters for optical connections from LVSG to PMS. PMS i/o units shall be rail-mounted remote i/o units and may be by the circuit breaker manufacturer or a reputable third party.
 - (ii) Provide Data Concentrators within LVSG's to combine data interfaces of multiple devices into a single interface, when required to suit the capabilities of the plant control system architecture;
 - (iii) Wiring internal to LVSG: Connect LVSG devices to Data Concentrators within the LVSG; Connect Data Concentrators to field control network Ethernet Switch in RIO panel. If the LVSG internal network is a daisy-chain, it shall be arranged so that the withdrawal of one of more functional units only interrupts communications to the withdrawn units;
 - (iv) Data Concentrator (Fieldbus Gateway) shall be as follows:
 - 1) Data concentrators shall be intelligent devices which actively concentrate communications to reduce the communications overhead and burden on upstream PLC's;
 - 2) Factory installed within a compartment of the LVSG which is dedicated to communications hardware, powered from UPS-derived supplies; and
 - 3) Typically a product of the LVSG manufacturer.
 - (v) PMS Communications System Interface: Each power meter, protection relay, and PMS i/o unit shall be capable of communicating to the Power Monitoring System (PMS) over an Ethernet network:
 - 1) Media Connection: Copper (internally) and Fibre optic (externally);
 - 2) Protocol: ModBus TCP/IP or other protocol to suit Power Monitoring System. Refer to Particular Specifications P6 - ICA;
 - 3) Network Addressing: Co-ordinated through S.O., entered and verified after units are in place and operational;
 - 4) Data Registers: As a minimum, provide the following information as data which can be queried by the host:
 - All items listed for display; and
 - All alarms.
- (n) All power measurement and power quality monitoring data shall be sent to host computer / SCADA / PMS / EMS system as per Particular Specifications and Drawings.
- (o) Mimic Bus
 - (i) One-line displaying bus arrangement, circuit protective devices, and branch circuit extensions;
 - (ii) Breaker Status Indicating Lights: LED type, green for breaker closed, red for breaker open;

- (iii) Buses: 10 mm wide, with red coloured strips;
- (iv) Devices: 6 mm wide, with red coloured strips;
- (v) Strips: Self-sticking plastic tape, applied to face of switchgear; and
- (vi) Nameplates to identify incoming lines, protective devices, and feeder circuits.
- (p) Equipment Identification
 - (i) Master Nameplate
 - 1) Deep-etched aluminium with manufacturer's name and model number; and
 - 2) Riveted to main vertical section.
 - (ii) Section Identification
 - 1) Stamped or engraved metallic, riveted to each vertical section;
 - 2) Serial number, bus rating, and section reference number; and
 - 3) Size: Manufacturer's standard.
 - (iii) Nameplate
 - 1) Engraved, acrylic or phenolic for each circuit breaker cubicle and door mounted device;
 - 2) White with black block type characters;
 - 3) Character Height: 6 mm;
 - 4) Size: As required for required inscription;
 - 5) Inscriptions: As shown on single-line diagram;
 - 6) Blank plates for future spaces; and
 - 7) Attachment Screws: Stainless steel panhead.
 - (iv) Cubicle Labels
 - 1) Non-metallic, applied inside each cubicle compartment;
 - 2) Device serial number, rating, and description; and
 - 3) Size: As required.
 - (v) Metering Instruments: Metre type identified on metre face below pointer or dial.
 - (vi) Control Switches: Deep-etched, aluminium escutcheon plate.
 - (vii) Relays and Devices
 - 1) Stamped metallic, riveted to instrument case; and
 - 2) Manufacturer's name, model number, relay type, and rating data.
 - (viii) Switchgear Signs
 - 1) Two signs each on front and back of switchgear;
 - 2) Size: As required;
 - 3) Engraved, acrylic or phenolic;

- 4) Colour: Red with white;
 - 5) Inscription: DANGER HIGH VOLTAGE / KEEP OUT in English and DANGER in Chinese, Tamil and Malay;
 - 6) Characters: 50 mm high; and
 - 7) Attachment: Four rivets each sign.
- (q) Interlocks
- (i) Key Interlocks
 - 1) Mechanical lock cylinder, located as shown on the Drawings;
 - 2) Keys to be captive when breakers are closed; and
 - 3) Two keys available for each group of three locks.
 - (ii) Electrical & Mechanical Interlocks
 - 1) Contractor shall design and develop necessary electrical and mechanical interlocks for safe operation of the electrical system;
 - 2) Contractor shall design and develop interlocking system such that two sources are not paralleled in any operating scenario. Necessary signals and interlocks shall be built into the ATC system either for operating in Auto or Manual mode; and
 - 3) All necessary CT, PT, control and interlocking cabling shall be provided between upstream HVSG's, Transformers and other LVSG's for implementation of interlocking system.
- (r) Testing
- (i) General
 - 1) Test all switchgear, including all components in accordance with IEC 61439; and
 - 2) Testing to be in accordance with established ISO 9001 procedures.
 - (ii) Factory Inspections
 - 1) Inspect for required construction, electrical connection, and intended function. All Source Quality Control and Factory Acceptance Tests (FAT) shall be witnessed by S.O.;
 - 2) Notify S.O. at least 2 weeks prior to any testing to allow S.O. to witness tests; and
 - 3) Completely assemble LVSG at the factory, and perform following tests at manufacturer's shop in accordance with IEC 61439:
 - Carry out routine test as part of FAT (Factory Acceptance Test) in accordance with IEC Routine Tests on all LVSG;
 - Conformance tests to demonstrate conformance of the equipment to applicable standards; and
 - Production tests to check the quality and uniformity of the workmanship and materials used in the manufacturer of the equipment.

G5.34.3 Execution

- (a) Installation
 - (i) Install equipment in accordance with manufacturer's instructions and recommendations;
 - (ii) The switchboards shall sit on concrete base of not less than 75 mm height on all four sides;
 - (iii) Secure equipment to mounting pads with anchor bolts;
 - (iv) Install equipment plumb and in longitudinal alignment with pad or wall; and
 - (v) Co-ordinate terminal connections with installation of secondary feeders.
- (b) Field Testing
 - (i) In accordance with General Specification G5.51 – Electrical Testing.
- (c) Manufacturer's Services
 - (i) Inspection, Commissioning and Site Adjustment: In addition to the requirements of General Specification G2.25 – Manufacturers' Services. An authorised service representative of the manufacturer shall visit Site as necessary to perform the following services:
 - 1) Verify proper installation of the equipment;
 - 2) Inspection, checking and adjusting the equipment; and
 - 3) Commissioning and field testing of the switchgear.
 - (ii) Training of operating Personnel: The authorised service representative or a training instructor shall train the operating personnel in all aspects of drive operation and maintenance, including step-by-step troubleshooting procedures with necessary test equipment and in accordance with General Specification G2.27 – Training, Advisory and Technical Support.
 - (iii) Provide Manufacturer's Certificate of Proper Installation.

END OF SECTION